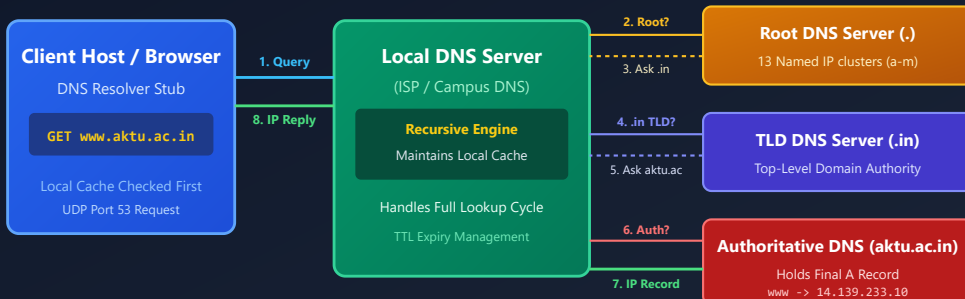


Module 01: Domain Name System (DNS) Architecture

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DOMAIN NAME SYSTEM (DNS) RESOLUTION ARCHITECTURE

Hierarchical Tree, Recursive vs Iterative Resolution, Resource Records & Caching



DNS PROTOCOL SPECIFICATIONS & RESOURCE RECORD (RR) ARCHITECTURE

1. Domain Tree Hierarchy

- Root Domain: Denoted by dot (.)
- TLD: Generic (.com, .edu, .org, .gov)
- ccTLD: Country Code (.in, .us, .uk, .jp)
- Second-Level / Subdomains: aktu.ac.in

2. Resolution Types

- Recursive Query: Server does full lookup and returns final answer to client.
- Iterative Query: Server returns referral IP pointer to next lower server.

3. Key Resource Records (RR)

- A: Hostname -> IPv4 (32-bit)
- AAAA: Hostname -> IPv6 (128-bit)
- CNAME: Alias canonical domain name
- MX: Mail exchange server | NS: Nameserver

1. Domain Name System (DNS) Overview

DNS translates human-readable hostnames into machine-routable IP addresses using a distributed, hierarchical database.

Protocol Standards & Port Assignment

Transport Protocol: UDP Port 53 for standard interactive queries (≤ 512 bytes). TCP Port 53 for zone transfers and extended EDNS0 responses (> 512 bytes).

2. Domain Name Space Divisions

Division	Scope & Syntax	Standard Examples
Generic Domains (gTLD)	Categorizes registered institutions by organizational nature	.com (Commercial), .edu (Educational), .org (Non-profit), .gov (Government)
Country Domains (ccTLD)	Two-letter geographic national country codes	.in (India), .us (United States), .uk (United Kingdom), .jp (Japan)
Inverse Domains (PTR)	Reverse mapping: Translates known IP address back into hostname	in-addr.arpa special domain zone for IPv4 reverse resolution

3. Recursive vs Iterative Query Resolution

Criterion	Recursive Resolution	Iterative Resolution
Client Expectation	Demands definitive final answer or explicit failure error	Accepts best-effort referral pointer to subordinate server
Processing Workload	Heavy computation borne by the interrogated server	Distributed across querying client / recursive ISP resolver
Topology Placement	Client Stub Resolver $\leftrightarrow$ Local ISP DNS Server	Local ISP DNS Server $\leftrightarrow$ Root $\rightarrow$ TLD $\rightarrow$ Authoritative Servers

4. Core Resource Record (RR) Types

Format: <Domain_Name, Value, Type, TTL>

- **A Record:** Resolves hostname to 32-bit IPv4 address.

- **AAAA Record:** Resolves hostname to 128-bit IPv6 address.
- **CNAME Record:** Canonical alias pointing alias domain to master domain.
- **MX Record:** Identifies primary and backup mail exchange routing servers.
- **NS Record:** Delegates DNS zone authority to specified nameservers.